



Task 1: Neural Networks

Team: 40

2022170136

حنين هانى عبد المنعم

2022170149

رحمه محمد خطاب حامد

2022170165

زياد خالد رشاد شرف

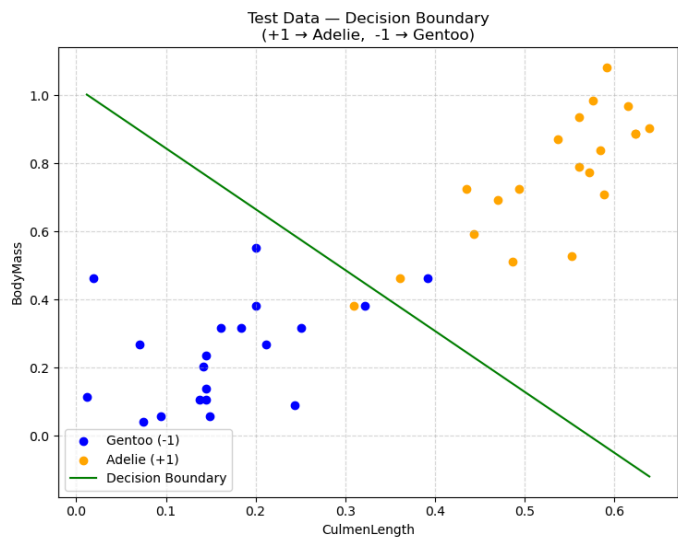
2022170242

عبدالرحمن مصطفى حسن

2022170393

محمد وائل مروان البكري

Perceptron Model

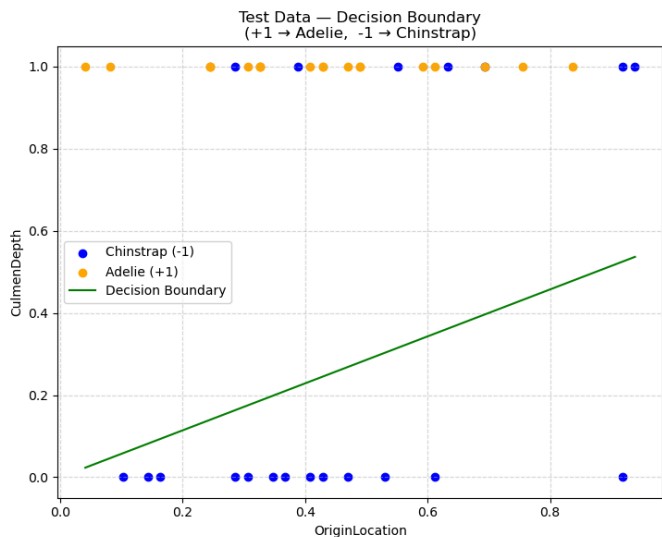


Confusion Matrix

	Pred Adelie	Pred Gentoo
Actual Adelie	19 (TP)	1 (FN)
Actual Gentoo	1 (FP)	19 (TN)

Parameter	Value
Features Used	CulmenLength, BodyMass
Classes Used	Gentoo, Adelie
Learning Rate	0.05
Epochs	100
Bias Included	Yes
Bias Value	-0.5000
Weights	[0.8737, 0.4890]
Accuracy	95.00%
Algorithm	Perceptron

1. The model made the decision boundary successfully, but two points were hard to classify.

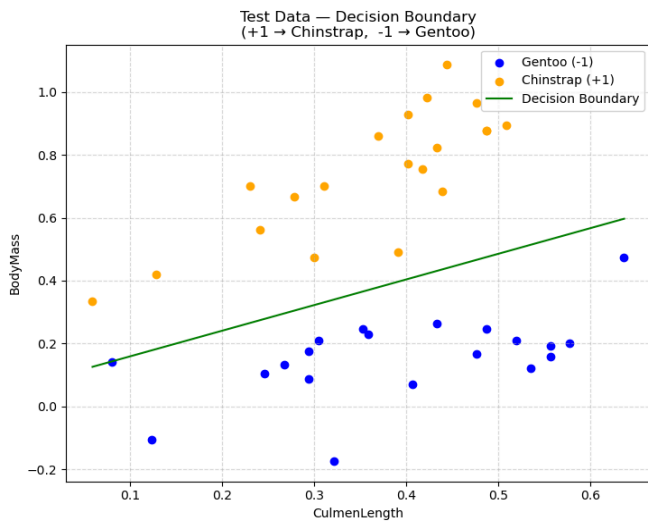


Confusion Matrix

	Pred Adelie	Pred Chinstrap
Actual Adelie	20 (TP)	0 (FN)
Actual Chinstrap	7 (FP)	13 (TN)

Parameter	Value
Features Used	OriginLocation, CulmenDepth
Classes Used	Chinstrap, Adelie
Learning Rate	0.001
Epochs	200
Bias Included	No
Bias Value	—
Weights	[-0.0009, 0.0015]
Accuracy	82.50%
Algorithm	Perceptron

2. The model made the decision boundary, but the data points were hard to classify, and the features didn't discriminate between the classes.

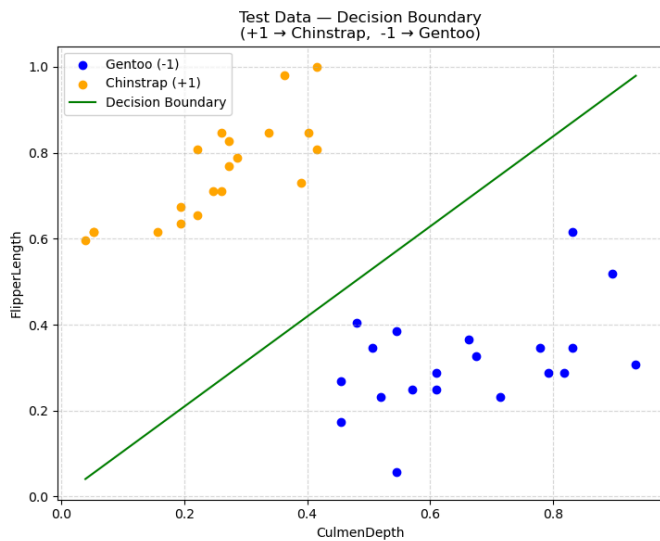


Confusion Matrix

	Pred Chinstrap	Pred Gentoo
Actual Chinstrap	20 TP	0 FN
Actual Gentoo	0 FP	20 TN

Parameter	Value
Features Used	CulmenLength, BodyMass
Classes Used	Gentoo, Chinstrap
Learning Rate	0.05
Epochs	100
Bias Included	Yes
Bias Value	-0.1000
Weights	[-1.0462, 1.2828]
Accuracy	100.00%
Algorithm	Perceptron

3. The model made the decision boundary successfully and classified all the points correctly (with one point hardly classified).

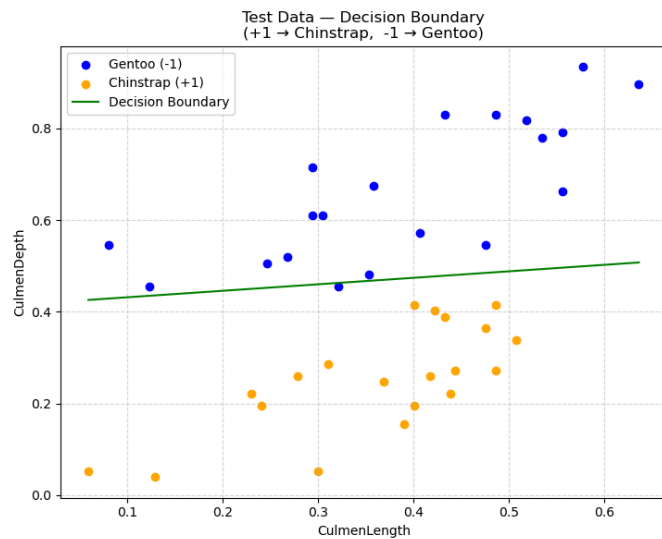


Confusion Matrix

	Pred Chinstrap	Pred Gentoo
Actual Chinstrap	20 TP	0 FN
Actual Gentoo	0 FP	20 TN

Parameter	Value
Features Used	CulmenDepth, FlipperLength
Classes Used	Gentoo, Chinstrap
Learning Rate	0.001
Epochs	400
Bias Included	No
Bias Value	—
Weights	[-0.0012, 0.0011]
Accuracy	100.00%
Algorithm	Perceptron

4. The model made the decision boundary successfully and classified all the points correctly.

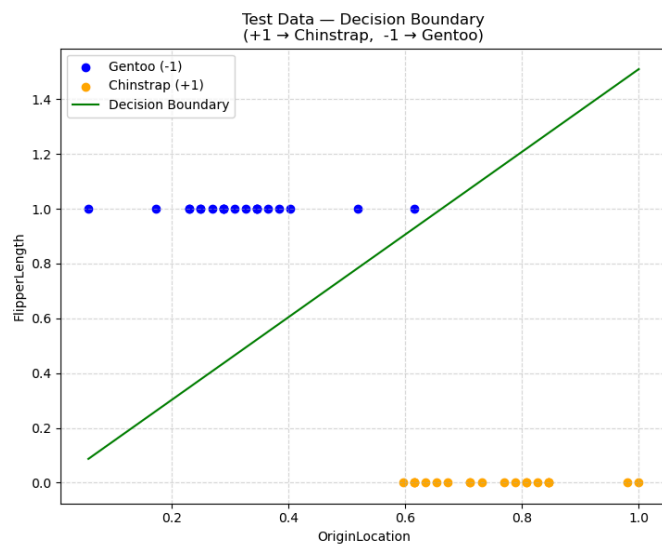


Confusion Matrix

	Pred Chinstrap	Pred Gentoo
Actual Chinstrap	20 TP	0 FN
Actual Gentoo	1 FP	19 TN

Parameter	Value
Features Used	CulmenLength, CulmenDepth
Classes Used	Gentoo, Chinstrap
Learning Rate	0.01
Epochs	100
Bias Included	Yes
Bias Value	0.2000
Weights	[0.0678, -0.4789]
Accuracy	97.50%
Algorithm	Perceptron

5. There was one point that was wrongly classified.



Confusion Matrix

	Pred Chinstrap	Pred Gentoo
Actual Chinstrap	20 TP	0 FN
Actual Gentoo	0 FP	20 TN

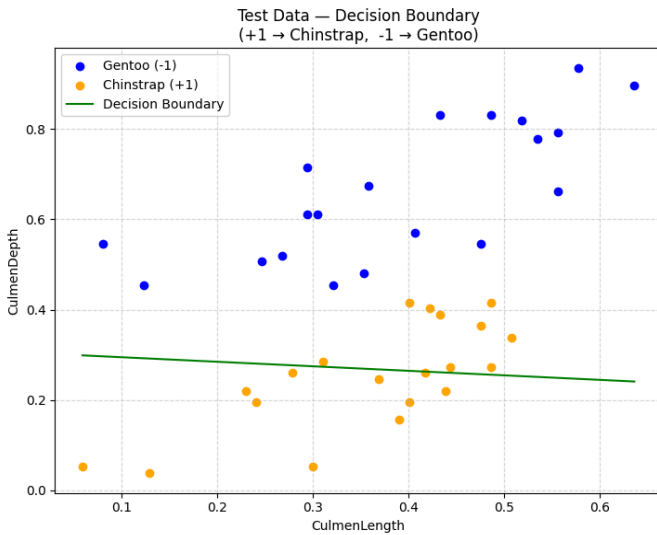
Parameter	Value
Features Used	OriginLocation, FlipperLength
Classes Used	Gentoo, Chinstrap
Learning Rate	0.01
Epochs	100
Bias Included	No
Bias Value	—
Weights	[0.3504, -0.2321]
Accuracy	100.00%
Algorithm	Perceptron

6. The model made the decision boundary successfully and classified all the points correctly.

Features with the highest accuracy

- CulmenLength and BodyMass (Fig. 3)
- CulmenDepth and FlipperLength (Fig. 4)
- OriginLocation and FlipperLength (Fig. 6)

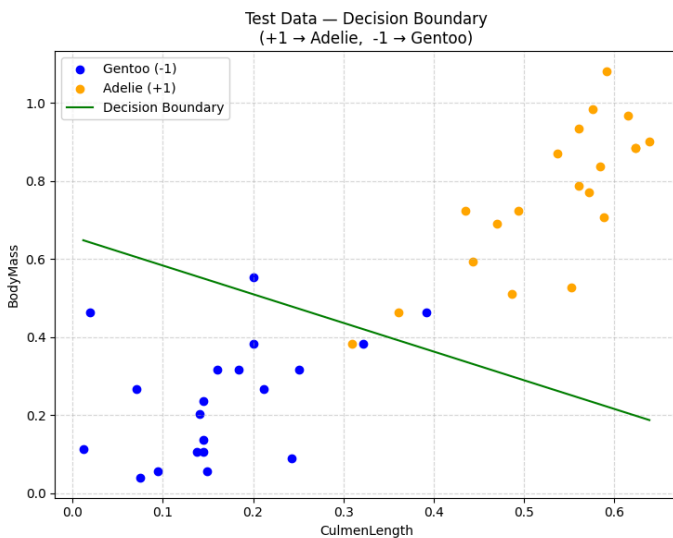
Adaline Model



Confusion Matrix	
Actual Chinstrap	11
	9
Actual Gentoo	0
	20
Pred Chinstrap Pred Gentoo	

Parameter	Value
Features Used	CulmenLength, CulmenDepth
Classes Used	Gentoo, Chinstrap
Learning Rate	0.01
Epochs	100
Bias Included	Yes
Bias Value	0.1218
Weights	[-0.0400, -0.3998]
Accuracy	77.50%
Algorithm	Adaline
MSE Threshold	0.01

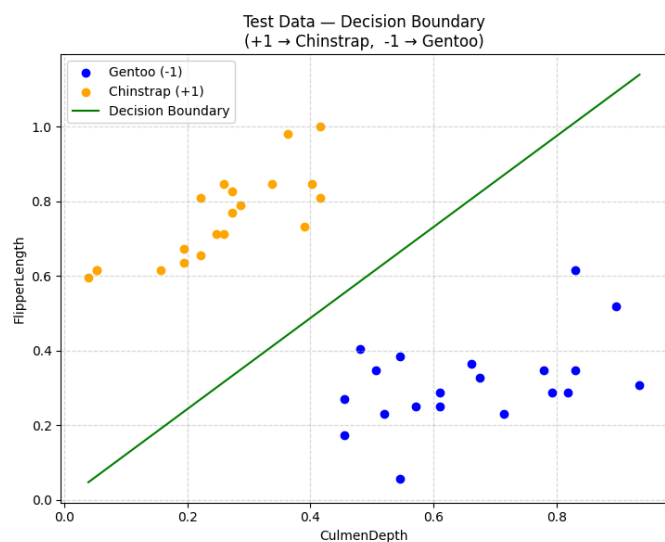
1. The model failed to classify the data correctly, and the model had bad performance here.



Confusion Matrix	
Actual Adelie	19
	1
Actual Gentoo	2
	18
Pred Adelie Pred Gentoo	

Parameter	Value
Features Used	CulmenLength, BodyMass
Classes Used	Gentoo, Adelie
Learning Rate	0.05
Epochs	100
Bias Included	Yes
Bias Value	-0.7342
Weights	[0.8205, 1.1172]
Accuracy	92.50%
Algorithm	Adaline
MSE Threshold	0.005

2. The model classified the data, but three points were hard to classify.

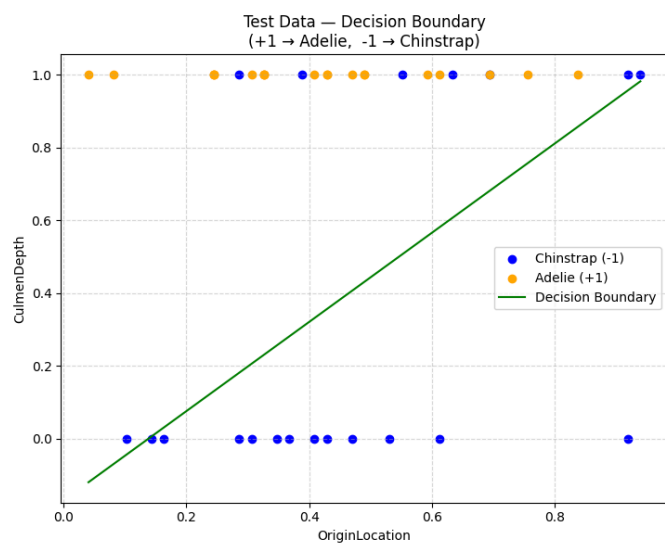


Confusion Matrix

	Pred Chinstrap	Pred Gentoo
Actual Chinstrap	20 TP	0 FN
Actual Gentoo	0 FP	20 TN

Parameter	Value
Features Used	CulmenDepth, FlipperLength
Classes Used	Gentoo, Chinstrap
Learning Rate	0.001
Epochs	400
Bias Included	No
Bias Value	—
Weights	[-0.1820, 0.1494]
Accuracy	100.00%
Algorithm	Adaline
MSE Threshold	0.001

3. The data is correctly classified.

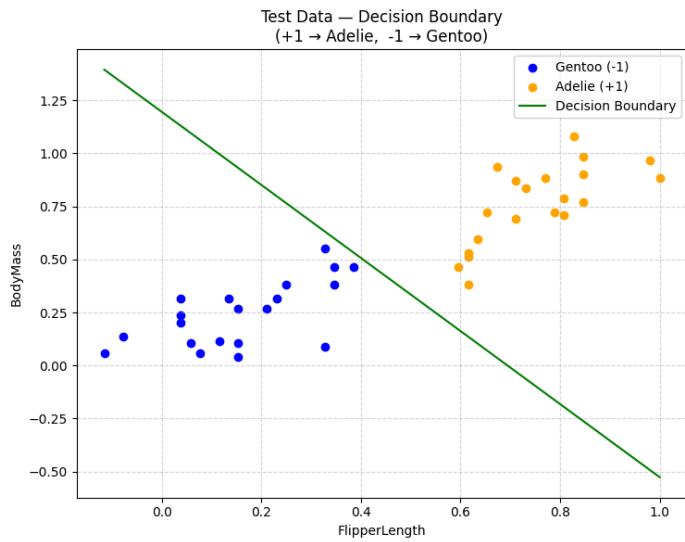


Confusion Matrix

	Pred Adelie	Pred Chinstrap
Actual Adelie	20 TP	0 FN
Actual Chinstrap	7 FP	13 TN

Parameter	Value
Features Used	OriginLocation, CulmenDepth
Classes Used	Chinstrap, Adelie
Learning Rate	0.001
Epochs	300
Bias Included	Yes
Bias Value	-0.0196
Weights	[0.1410, -0.1149]
Accuracy	82.50%
Algorithm	Adaline
MSE Threshold	0.005

4. The data is difficult to classify, and the model incorrectly classified one easy point.

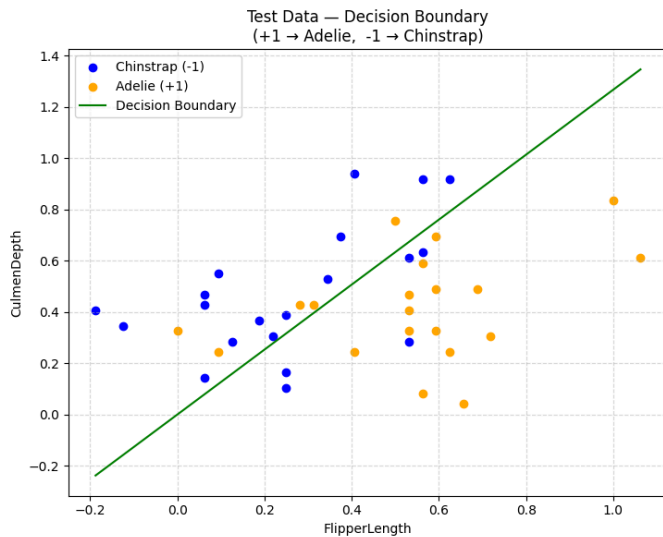


Confusion Matrix

	Pred Adelie	Pred Gentoo
Actual Adelie	20 TP	0 FN
Actual Gentoo	0 FP	20 TN

Parameter	Value
Features Used	FlipperLength, BodyMass
Classes Used	Gentoo, Adelie
Learning Rate	0.05
Epochs	200
Bias Included	Yes
Bias Value	-1.2593
Weights	[1.8140, 1.0527]
Accuracy	100.00%
Algorithm	Adaline
MSE Threshold	0.001

5. The data is correctly classified.



Confusion Matrix

	Pred Adelie	Pred Chinstrap
Actual Adelie	15 TP	5 FN
Actual Chinstrap	5 FP	15 TN

Parameter	Value
Features Used	FlipperLength, CulmenDepth
Classes Used	Chinstrap, Adelie
Learning Rate	0.03
Epochs	250
Bias Included	No
Bias Value	—
Weights	[1.0259, -0.8093]
Accuracy	75.00%
Algorithm	Adaline
MSE Threshold	0.003

6. The data is very hard to classify, and the features didn't discriminate between the classes.

Features with the highest accuracy

- CulmenDepth and FlipperLength (Fig. 3)
- BodyMass and FlipperLength (Fig. 5)